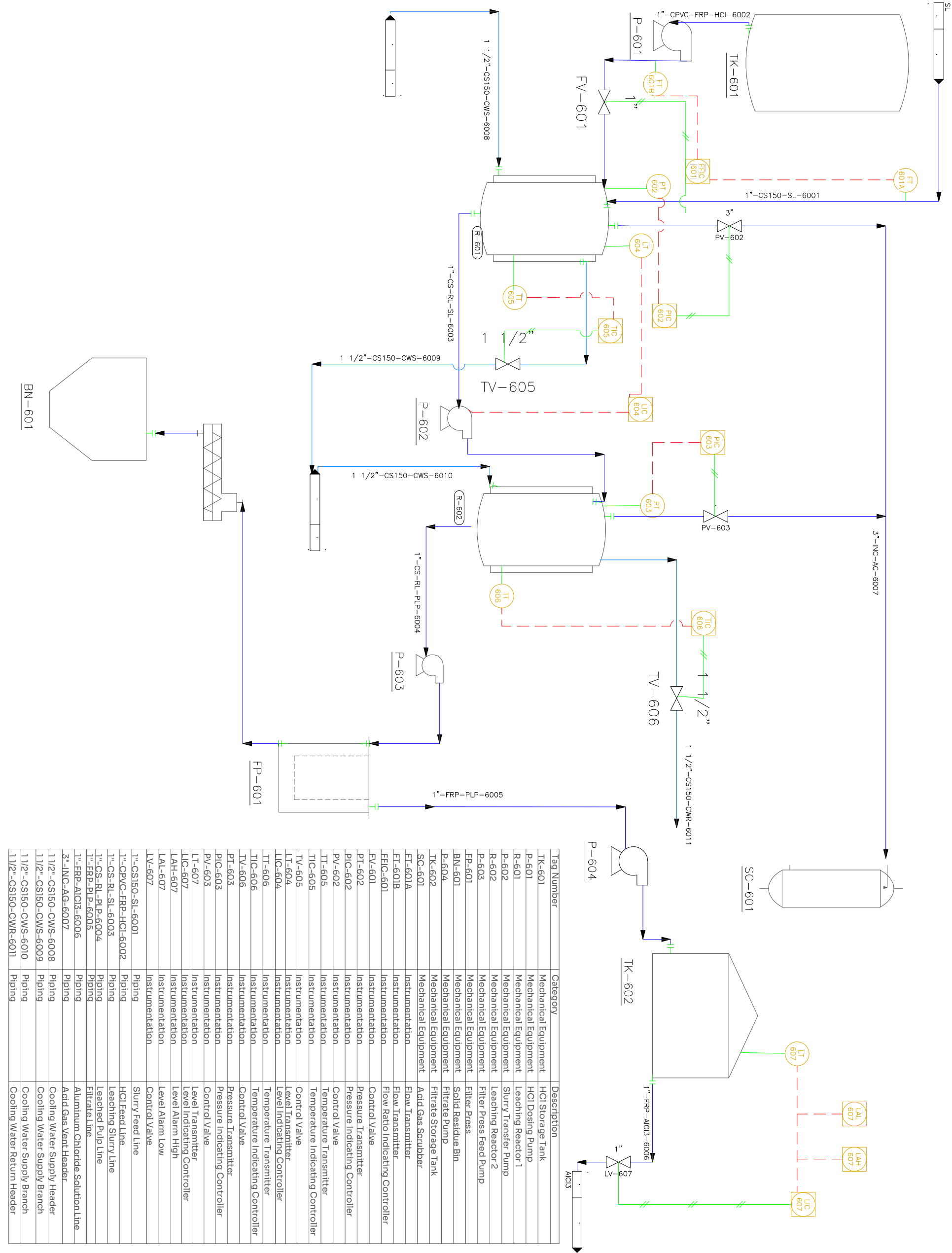




Tag Number	Category	Description
TK-601	Mechanical Equipment	HCl Storage Tank
P-601	Mechanical Equipment	HCl Dosing Pump
R-601	Mechanical Equipment	Leaching Reactor 1
P-602	Mechanical Equipment	Slurry Transfer Pump
R-602	Mechanical Equipment	Leaching Reactor 2
P-603	Mechanical Equipment	Filter Press Feed Pump
FP-601	Mechanical Equipment	Filter Press
BN-601	Mechanical Equipment	Solid Residue Bin
P-604	Mechanical Equipment	Filtrate Pump
TK-602	Mechanical Equipment	Filtrate Storage Tank
SC-601	Mechanical Equipment	Acid Gas Scrubber
FT-601A	Instrumentation	Flow Transmitter
FT-601B	Instrumentation	Flow Transmitter
FFIC-601	Instrumentation	Flow Ratio Indicating Controller
FV-601	Instrumentation	Control Valve
PT-602	Instrumentation	Pressure Transmitter
PI-602	Instrumentation	Pressure Indicating Controller
PV-602	Instrumentation	Control Valve
TT-602	Instrumentation	Temperature Transmitter
TIC-602	Instrumentation	Temperature Indicating Controller
TV-602	Instrumentation	Control Valve
TV-605	Instrumentation	Control Valve
TV-606	Instrumentation	Control Valve
PI-603	Instrumentation	Pressure Transmitter
PV-603	Instrumentation	Pressure Indicating Controller
LT-607	Instrumentation	Level Transmitter
LIC-607	Instrumentation	Level Indicating Controller
LAL-607	Instrumentation	Level Alarm High
LV-607	Instrumentation	Level Alarm Low
LV-607	Instrumentation	Control Valve
1"-CS150-SL-6001	Piping	Slurry Feed Line
1"-CPVC-FRP-HCl-6002	Piping	HCl Feed Line
1"-CS-RL-SL-6003	Piping	Leaching Slurry Line
1"-CS-RL-PLP-6004	Piping	Leached Pulp Line
1"-FRP-PLP-6005	Piping	Filtrate Line
1"-FRP-AIC3-6006	Piping	Aluminum Chloride Solution Line
3"-INC-AG-6007	Piping	Acid Gas Vent Header
1 1/2"-CS150-CWS-6008	Piping	Cooling Water Supply Header
1 1/2"-CS150-CWS-6009	Piping	Cooling Water Supply Branch
1 1/2"-CS150-CWS-6010	Piping	Cooling Water Supply Branch
1 1/2"-CS150-CWR-6011	Piping	Cooling Water Return Header

General Notes

- All pipes, fittings, and valves shall be designed in accordance with ASME B31.3 process piping standards.
- All instrumentation symbols and nomenclature conform to ISA 5.1 standards.
- Fall-safe actions for all pneumatic control valves (fail-open/fail-closed) shall be determined during the HAZOP review phase.
- Strict Material Specifications: Due to the combination of extreme corrosivity and abrasion, piping materials must be strictly adhered to: Rubber-Lined Carbon Steel (CS-RL) for leached slurry, PVC/FRP for Hydrochloric Acid (HCl), and Inconel (INC) for hot acid off-gas lines.
- Exothermic Reaction Interlock: The acid leaching process is highly exothermic. The HCl dosing pump must be hardwired/interlocked to automatically shut down upon loss of cooling water flow (CWS-6008/6010) to the reactor jackets to prevent runaway temperatures.
- The acid gas (AG) off-gas header routing to the scrubber (SC-601) handles highly volatile and toxic HCl fumes. The scrubber must be operational and drawing draft before any HCl dosing into the reactors can commence. All slurry transfer lines must maintain a minimum suspension velocity to prevent bauxite settling and must be equipped with dedicated flush-water tie-ins for line clearing during shutdowns.
-

No.	Revision/Issue	Date

Drawing Name

Area600
Leaching &
Solid-Liquid
Separation

Project Name and Address

PAC_P&ID
Final Year Project
PMB, UPO, KNUST, Ghana
Kumasi Ashanti Region

Area	600	Stamp
Drawing Number	6	
Author	GROUP 7B	